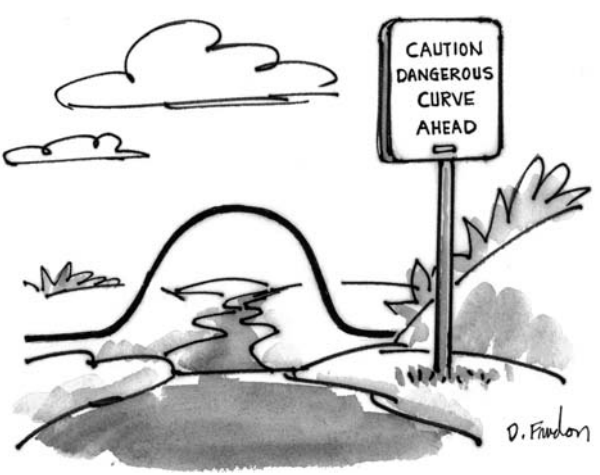
5

The Normal Approximation for Data

1. THE NORMAL CURVE

The normal curve was discovered around 1720 by Abraham de Moivre, while he was developing the mathematics of chance. (His work will be discussed again in parts IV and V.) Around 1870, the Belgian mathematician Adolph Quetelet had the idea of using the curve as an ideal histogram, to which histograms for data could be compared.

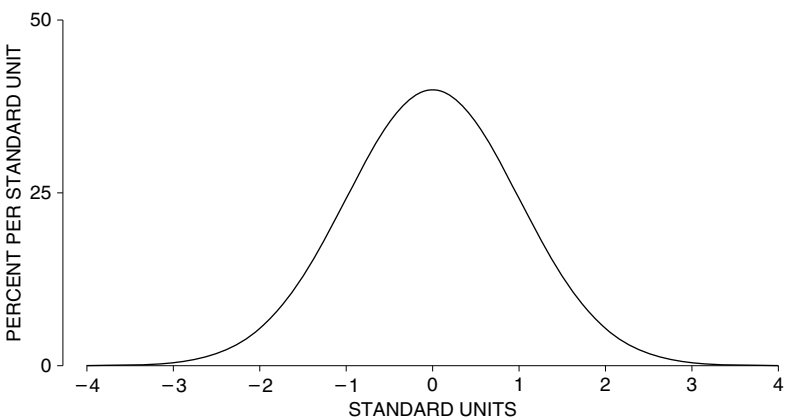


The normal curve has a formidable-looking equation:

$$y = \frac{100\%}{\sqrt{2\pi}} e^{-x^2/2}, \quad \text{where } e = 2.71828\ldots.$$

This equation involves three of the most famous numbers in the history of mathematics: $\sqrt{2}$, π, and e. This is just to show off a little. You will find it is easy to work with the normal curve through diagrams and tables, without ever using the equation. A graph of the curve is shown in figure 1.

Figure 1. The normal curve.



Several features of this graph will be important. First, the graph is symmetric about 0: the part of the curve to the right of 0 is a mirror image of the part to the left. Next, the total area under the curve equals 100%. (Areas come out in percent, because the vertical axis uses the density scale.) Finally, the curve is always above the horizontal axis. It appears to stop between 3 and 4, but that's only because the curve gets so low there. Only about 6/100,000 of the area is outside the interval from −4 to 4.

It will be helpful to find areas under the normal curve between specified values. For instance,

- the area under the normal curve between −1 and +1 is about 68%;
- the area under the normal curve between −2 and +2 is about 95%;
- the area under the normal curve between −3 and +3 is about 99.7%.

Finding these areas is a matter of looking things up in a table, or pushing a button on the right kind of calculator; the table will be explained in section 2.

Many histograms for data are similar in shape to the normal curve, provided they are drawn to the same scale. Making the horizontal scales match up involves *standard units*.[1]

> A value is converted to standard units by seeing how many SDs it is above or below the average.

Values above the average are given a plus sign; values below the average get a minus sign. The horizontal axis of figure 1 is in standard units.

For instance, take the women age 18 and over in the HANES5 sample. Their average height was 63.5 inches; the SD was 3 inches. One of these women was 69.5 inches tall. What was her height in standard units? Our subject was 6 inches taller than average, and 6 inches is 2 SDs. In standard units, her height was +2.

Example 1. For women age 18 and over in the HANES5 sample—

(a) Convert the following to standard units:
(i) 66.5 inches (ii) 57.5 inches (iii) 64 inches (iv) 63.5 inches

(b) Find the height which is −1.2 in standard units.

Solution. Part (a). For (i), 66.5 inches is 3 inches above the average. That is 1 SD above the average. In standard units, 66.5 inches is +1. For (ii), 57.5 inches is 6 inches below the average. That is 2 SDs below average. In standard units, 57.5 inches is −2. For (iii), 64 inches is 0.5 inches above average. That is $0.5/3 \approx 0.17$ SDs. The answer is 0.17. For (iv), 63.5 inches is the average. So, 63.5 inches is 0 SDs away from average. The answer is 0. (Reminder: "≈" means "nearly equal.")

Part (b). The height is 1.2 SDs below the average, and 1.2×3 inches = 3.6 inches. The height is

$$63.5 \text{ inches} - 3.6 \text{ inches} = 59.9 \text{ inches}.$$

That is the answer.

Standard units are used in figure 2. In this figure, the histogram for the heights of the women age 18 and over in the HANES5 sample is compared to the normal curve. The horizontal axis for the histogram is in inches; the horizontal axis for the normal curve is in standard units. The two match up as indicated in example 1. For instance, 66.5 inches is directly above +1, and 57.5 inches is directly above −2.

There are also two vertical axes in figure 2. The histogram is drawn relative to the inside one, in percent per inch. The normal curve is drawn relative to the outside one, in percent per standard unit. To see how the scales match up, take the top value on each axis: 60% per standard unit matches 20% per inch because there are 3 inches to the standard unit. Spreading 60% over an SD is the same as spreading 60% over 3 inches, and that comes to 20% per inch—

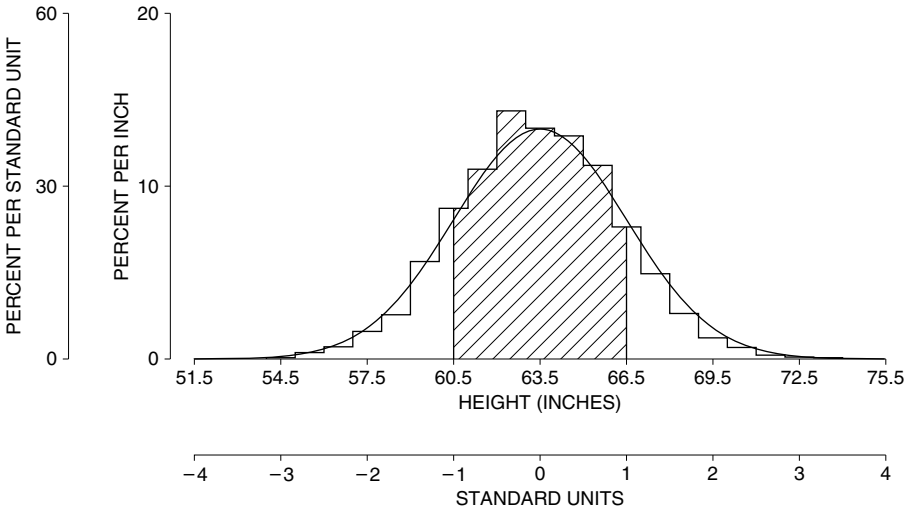
$$\begin{aligned} 60\% \text{ per standard unit} &= 60\% \text{ per 3 inches} \\ &= 60\% \div 3 \text{ inches} = 20\% \text{ per inch.} \end{aligned}$$

Similarly, 30% per standard unit matches 10% per inch. Any other pair of values can be dealt with in the same way.

The last chapter said that for many lists, roughly 68% of the entries are within one SD of average. This is the range

$$\text{average} - \text{SD} \quad \text{to} \quad \text{average} + \text{SD}.$$

Figure 2. A histogram for heights of women compared to the normal curve. The area under the histogram between 60.5 inches and 66.5 inches (the percentage of women within one SD of average with respect to height) is about equal to the area between −1 and +1 under the curve—68%.

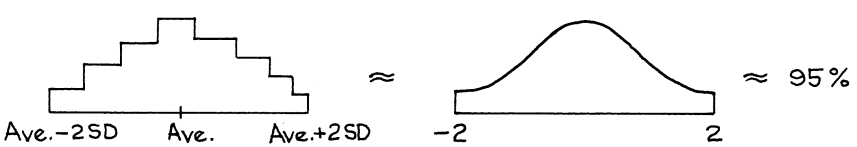


To see where the 68% comes from, look at figure 2. The percentage of women whose heights are within one SD of average equals the area under the histogram within one SD of average. This area is shaded in figure 2. The histogram follows the normal curve fairly well. Parts of it are higher than the curve, and parts of it are lower. But the highs balance out the lows. And the shaded area under the histogram is about the same as the area under the curve. The area under the normal curve between −1 and +1 is 68%. That is where the 68% comes from.

For many lists, roughly 95% of the entries are within 2 SDs of average. This is the range

$$\text{average} - 2\,\text{SDs} \quad \text{to} \quad \text{average} + 2\,\text{SDs}.$$

The reasoning is similar. If the histogram follows the normal curve, the area under the histogram will be about the same as the area under the curve. And the area under the curve between −2 and +2 is 95%:



The normal curve can be used to estimate the percentage of entries in an interval, as follows.[2] First, convert the interval to standard units; second, find the

corresponding area under the normal curve. The method for getting areas will be explained in section 2. Finally, section 3 will put the two steps together. The whole procedure is called the *normal approximation*. The approximation consists in replacing the original histogram by the normal curve before finding the area.

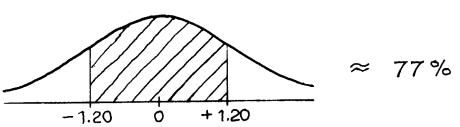
Exercise Set A

1. On a certain exam, the average of the scores was 50 and the SD was 10.
   (a) Convert each of the following scores to standard units: 60, 45, 75.
   (b) Find the scores which in standard units are: 0, +1.5, −2.8.
2. (a) Convert each entry on the following list to standard units (that is, using the average and SD of the list): 13, 9, 11, 7, 10.
   (b) Find the average and SD of the converted list.

The answers to these exercises are on p. A51.

2. FINDING AREAS UNDER THE NORMAL CURVE

At the end of the book, there is a table giving areas under the normal curve (p. A104). For example, to find the area under the normal curve between −1.20 and 1.20, go to 1.20 in the column marked z and read off the entry in the column marked *Area*. This is about 77%, so the area under the normal curve between −1.20 and 1.20 is about 77%.

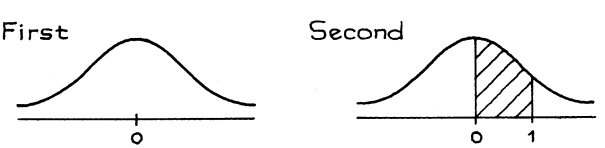


But you are also going to want to find other areas:

The method for finding such areas is indicated by example.

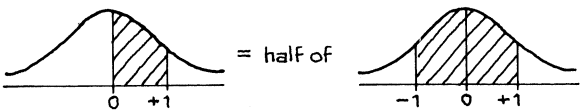
Example 2. Find the area between 0 and 1 under the normal curve.

Solution. First make a sketch of the normal curve, and then shade in the area to be found.



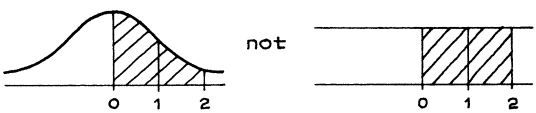
The table will give you the area between -1 and $+1$. This is about 68%. By symmetry, the area between 0 and 1 is half the area between -1 and $+1$, that is,

$$\frac{1}{2} \times 68\% = 34\%$$



Example 3. Find the area between 0 and 2 under the normal curve.

Solution. This isn't double the area between 0 and 1 because the normal curve isn't a rectangle.

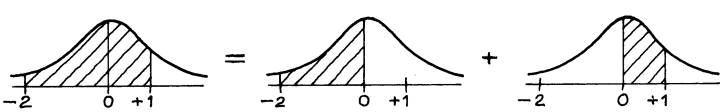


The procedure is the same as in example 2. The area between -2 and 2 can be found from the table. It is about 95%. The area between 0 and 2 is half that, by symmetry:

$$\frac{1}{2} \times 95\% \approx 48\%.$$

Example 4. Find the area between -2 and 1 under the normal curve.

Solution. The area between -2 and 1 can be broken down into two other areas—

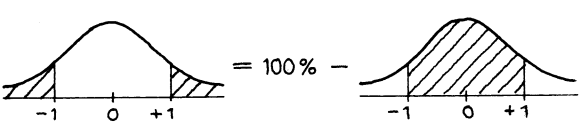


The area between -2 and 0 is the same as the area between 0 and 2, by symmetry, and is about 48% (example 3). The area between 0 and 1 is about 34% (example 2). The area between -2 and 1 is about

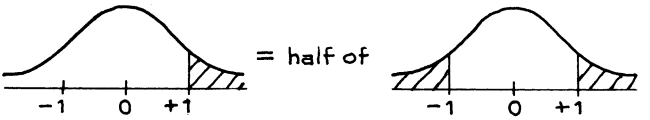
$$48\% + 34\% = 82\%.$$

Example 5. Find the area to the right of 1 under the normal curve.

Solution. The table gives the area between -1 and 1, which is 68%. The area outside this interval is 32%.

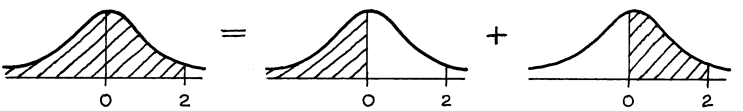


By symmetry, the area to the right of 1 is half this, or 16%.



Example 6. Find the area to the left of 2 under the normal curve.

Solution. The area to the left of 2 is the sum of the area to the left of 0, and the area between 0 and 2.



The area to the left of 0 is half the total area, by symmetry:

$$\frac{1}{2} \times 100\% = 50\%$$

The area between 0 and 2 is about 48%. The sum is $50\% + 48\% = 98\%$.

Example 7. Find the area between 1 and 2 under the normal curve.

Solution.

The area between -2 and 2 is about 95%; the area between -1 and 1 is about 68%. Half the difference is

$$\frac{1}{2} \times (95\% - 68\%) = \frac{1}{2} \times 27\% \approx 14\%.$$

There is no set procedure to use in solving this sort of problem. It is a matter of drawing pictures which relate the area you want to areas that can be read from the table.

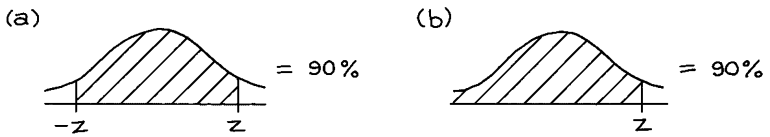
Exercise Set B

1. Find the area under the normal curve—

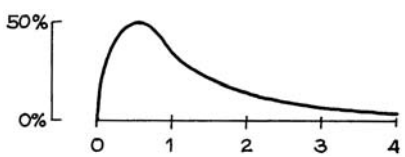
(a) to the right of 1.25	(b) to the left of -0.40
(c) to the left of 0.80	(d) between 0.40 and 1.30
(e) between -0.30 and 0.90	(f) outside -1.5 to 1.5

2. Fill in the blanks:
   (a) The area between ± _____ under the normal curve equals 68%.
   (b) The area between ± _____ under the normal curve equals 75%.

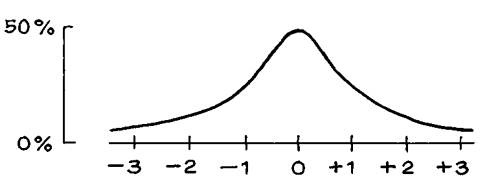
3. The normal curve is sketched below; solve for z.



4. A certain curve (not the normal) is sketched below. The total area under it is 100%, and the area between 0 and 1 is 39%.
   (a) If possible, find the area to the right of 1.
   (b) If possible, find the area between 0 and 0.5.



5. A certain curve (not the normal) is sketched below. It is symmetric around 0, and the total area under it is 100%. The area between −1 and 1 is 58%.
   (a) If possible, find the area between 0 and 1.
   (b) If possible, find the area to the right of 1.
   (c) If possible, find the area to the right of 2.



The answers to these exercises are on p. A51.

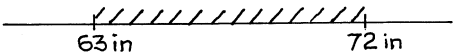
3. THE NORMAL APPROXIMATION FOR DATA

The method for the normal approximation will be explained here by example. The diagrams look so simple that you may not think they are worth drawing. However, it is easy to lose track of the area that is wanted. Please draw the diagrams.

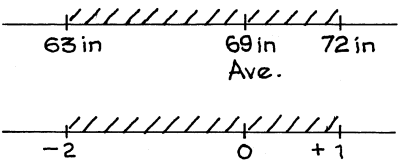
Example 8. The heights of the men age 18 and over in HANES5 averaged 69 inches; the SD was 3 inches. Use the normal curve to estimate the percentage of these men with heights between 63 inches and 72 inches.

Solution. The percentage is given by the area under the height histogram, between 63 inches and 72 inches.

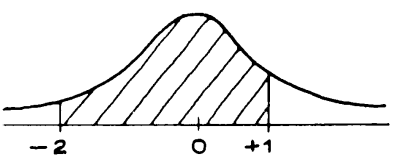
Step 1. Draw a number line and shade the interval.



Step 2. Mark the average on the line and convert to standard units.

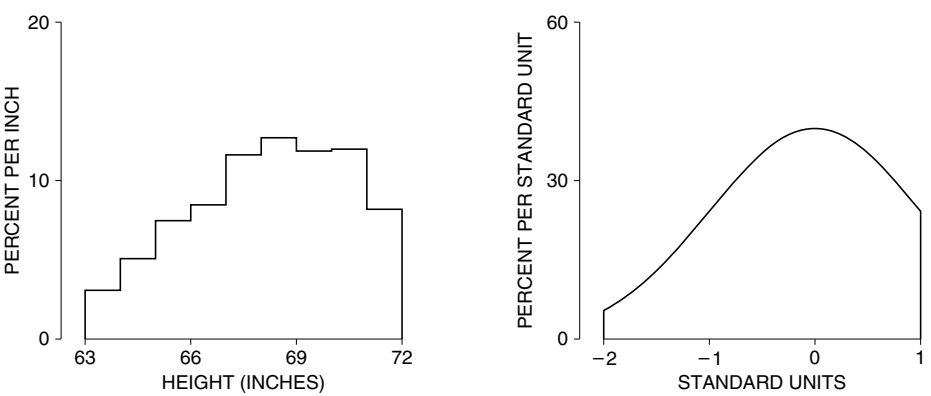


Step 3. Sketch in the normal curve, and find the area above the shaded standard-units interval obtained in step 2. The percentage is approximately equal to the shaded area, which is almost 82%.



Using the normal curve, we estimate that about 82% of the heights were between 63 inches and 72 inches. This is only an approximation, but it is pretty good: 81% of the men were in that range. Figure 3 shows the approximation.

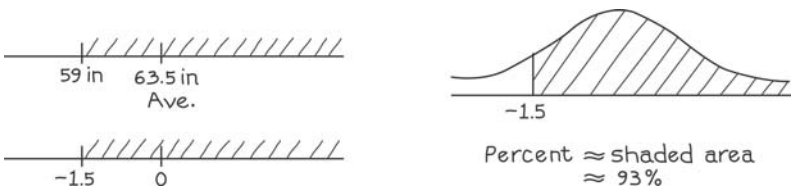
Figure 3. The normal approximation consists in replacing the original histogram by the normal curve before computing areas.



Example 9. The heights of the women age 18 and over in HANES5 averaged 63.5 inches; the SD was 3 inches. Use the normal curve to estimate the percentage with heights above 59 inches.

Solution. A height of 59 inches is 1.5 SDs below average:

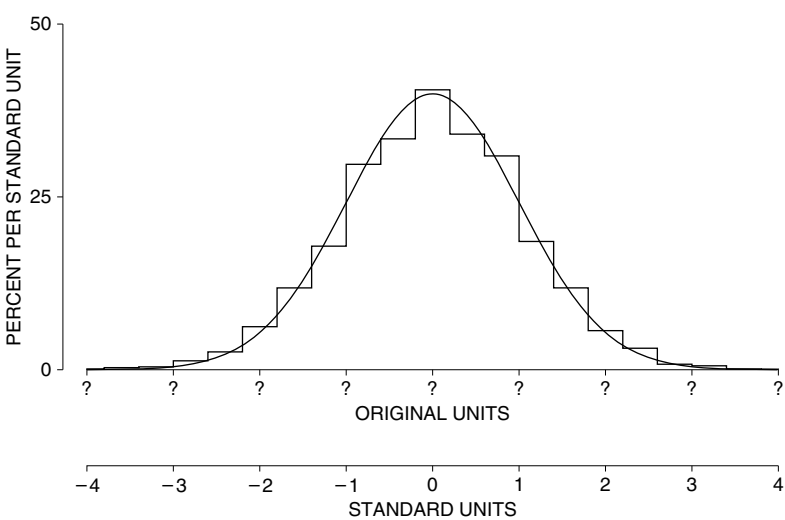
$$(59 - 63.5)/3 = -1.5.$$



Using the normal curve, we estimate that 93% of the women were more than 59 inches in height. This estimate is about right: 96% of the women were taller than 59 inches.

It is a remarkable fact that many histograms follow the normal curve. (The story continues in part V.) For such histograms, the average and SD are good summary statistics. If a histogram follows the normal curve, it looks something like the sketch in figure 4. The average pins down the center, and the SD gives the spread. That is nearly all there is to say about the histogram—if its shape is like the normal curve. Many other histograms, however, do not follow the normal curve. In such cases, the average and SD are poor summary statistics. More about this in the next section.

Figure 4. The average and SD. By locating the center and measuring the spread around the center, the average and SD summarize a histogram which follows the normal curve.



Exercise Set C

1. For the women age 18–24 in HANES2, the average height was about 64.3 inches; the SD was about 2.6 inches. Using the normal curve, estimate the percentage of women with heights—
   (a) below 66 inches.
   (b) between 60 inches and 66 inches.
   (c) above 72 inches.

2. In a law school class, the entering students averaged about 160 on the LSAT; the SD was about 8. The histogram of LSAT scores followed the normal curve reasonably well. (LSAT scores range from 120 to 180; among all test-takers, the average is around 150 and the SD is around 9.)
   (a) About what percentage of the class scored below 166?
   (b) One student was 0.5 SDs above average on the LSAT. About what percentage of the students had lower scores than he did?

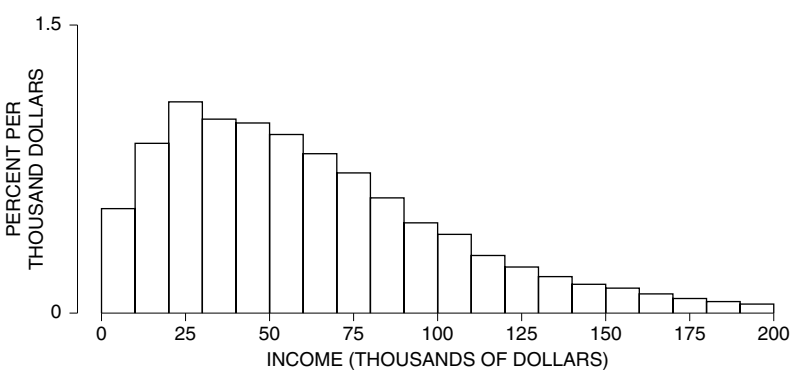
3. In figure 2 (p. 81), the percentage of women with heights between 61 inches and 66 inches is exactly equal to the area between 61 inches and 66 inches under the _____ and approximately equal to the area under the _____. Options: normal curve, histogram.

The answers to these exercises are on pp. A51–52.

4. PERCENTILES

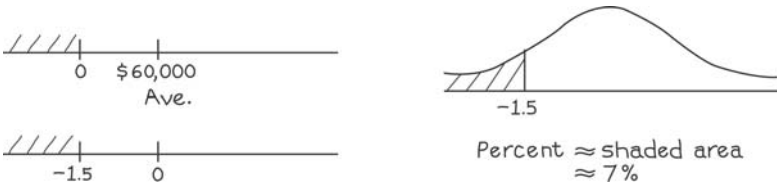
The average and SD can be used to summarize data following the normal curve. They are less satisfactory for other kinds of data. Take the distribution of family income in the U.S. in 2004, shown in figure 5.

Figure 5. Distribution of families by income: the U.S. in 2004.



Source: March 2005 Current Population Survey; CD-ROM supplied by the Bureau of the Census. Primary families.

The average income for the families in figure 5 was about \$60,000; the SD was about \$40,000.[3] So the normal approximation suggests that about 7% of these families had negative incomes:



The reason for this blunder: the histogram in figure 5 does not follow the normal curve at all well, it has a long right-hand tail. To summarize such histograms, statisticians often use *percentiles* (table 1).

Table 1. Selected percentiles for family income in the U.S. in 2004.

1	$0
10	$15,000
25	$29,000
50	$54,000
75	$90,000
90	$135,000
99	$430,000

Source: March 2005 Current Population Survey; CD-ROM supplied by the Bureau of the Census. Primary families.

The 1st percentile of the income distribution was $0, meaning that about 1% of the families had incomes of $0 or less, and about 99% had incomes above that level. (Mainly, the families with no income were retired or not working for some other reason.) The 10th percentile was $15,000: about 10% of the families had incomes below that level, and 90% were above. The 50th percentile is just the median (chapter 4).

By definition, the *interquartile range* equals

$$75\text{th percentile} - 25\text{th percentile}.$$

This is sometimes used as a measure of spread, when the distribution has a long tail. For table 1, the interquartile range is $61,000.

For reasons of their own, statisticians call de Moivre's curve "normal." This gives the impression that other curves are abnormal. Not so. Many histograms follow the normal curve very well, and many others—like the income histogram—do not. Later in the book, we will present a mathematical theory which helps explain when histograms should follow the normal curve.

Exercise Set D

1. Fill in the blanks, using the options below.
   (a) The percentage of families in table 1 with incomes below $90,000 was about _____.
   (b) About 25% of the families in table 1 had incomes below _____.
   (c) The percentage of families in table 1 with incomes between $15,000 and $125,000 was about _____.

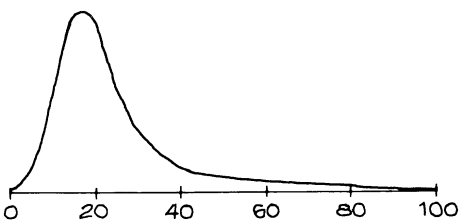
   5% 10% 25% 60% 75% 95% $29,000 $90,000

2. In 2004, a family with an income of \$9,000 was at the _____th percentile of the income distribution, while a family that made \$174,000 was at the _____th percentile. Options: 5, 95.

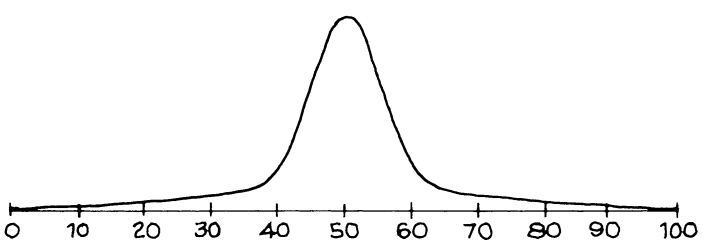
3. Is the 25th percentile for the distribution of family income in 1973 around \$7,000, \$10,000, or \$25,000? (See table 1 on p. 35.)

4. Skinfold thickness is used to measure body fat. A histogram for skinfold thickness is shown below; the units on the horizontal axis are millimeters (mm). The 25th percentile of skinfold thickness is _____25 mm. Fill in the blank, using one of the phrases below. Or can this be determined from the figure?

   quite a bit smaller than
   around
   quite a bit bigger than



5. A histogram is sketched below.
   (a) How is it different from the normal curve?
   (b) Is the interquartile range around 15, 25, or 50?



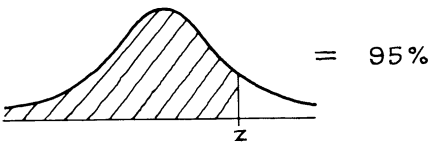
The answers to these exercises are on p. A52.

5. PERCENTILES AND THE NORMAL CURVE

When a histogram does follow the normal curve, the table can be used to estimate its percentiles. The method is indicated by example.

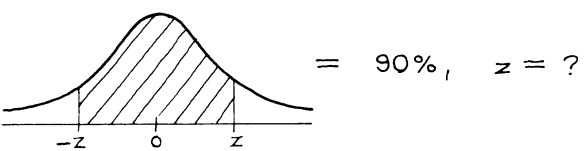
Example 10. Among all applicants to a certain university one year, the Math SAT scores averaged 535, the SD was 100, and the scores followed the normal curve. Estimate the 95th percentile of the score distribution.

Solution. This score is above average, by some number of SDs. We need to find that number, call it z. There is an equation for z:

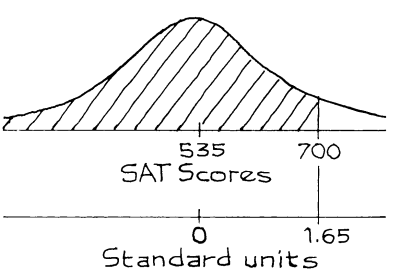


The normal table cannot be used directly, because it gives the area between $-z$ and z rather than the area to the left of z.

The area to the right of our z is 5%, so the area to the left of $-z$ is 5% too. Then the area between $-z$ and z must be $100\% - 5\% - 5\% = 90\%$.



From the table, $z \approx 1.65$. You have to score 1.65 SDs above average to be in the 95th percentile of the Math SAT. Translated back to points, this score is above average by $1.65 \times 100 = 165$ points. The 95th percentile of the score distribution is $535 + 165 = 700$.



The terminology is a little confusing. A *percentile* is a score: in example 10, the 95th percentile is a score of 700. A *percentile rank*, however, is a percent: if you score 700, your percentile rank is 95%. There is even a third way to say the same thing: a score of 700 puts you at the 95th percentile of the score distribution.

Exercise Set E

1. At the university in example 10, one applicant scored 750 on the Math SAT. She was at the _____ percentile of the score distribution.
2. For the university in example 10, estimate the 80th percentile of the Math SAT scores.
3. For Berkeley freshmen, the average GPA (grade point average) is around 3.0; the SD is about 0.5. The histogram follows the normal curve. Estimate the 30th percentile of the GPA distribution.

The answers to these exercises are on p. A52.

6. CHANGE OF SCALE

If you add the same number to every entry on a list, that number just gets added to the average; the SD does not change. (The deviations from the average do not change, because the added constant just cancels.) Furthermore, if you multiply every entry on a list by the same number, the average and the SD simply get multiplied by that number. There is one exception: if that constant multiplier is negative, wipe out its sign before applying it to the SD. Exercises 5–8 on p. 73 illustrated these ideas.

Example 11.

(a) Find the average and SD of the list 1, 3, 4, 5, 7.

(b) Take the list in part (a), multiply each entry by 3 and then add 7, to get the list 10, 16, 19, 22, 28. Find the average and SD of this new list.

Solution. Part (a). The average is 4. So the deviations from average are $-3, -1, 0, 1, 3$. The SD is 2.

Part (b). The average is $3 \times 4 + 7 = 19$, the SD is $3 \times 2 = 6$. (Of course, you can work these numbers out directly.)

Example 12. Convert the following lists to standard units:

(a) 1, 3, 4, 5, 7

(b) 10, 16, 19, 22, 28

(These are the two lists in the previous example.)

Solution. Part (a). The average is 4, and the deviations from average are $-3, -1, 0, 1, 3$. The SD is 2. Divide by 2 to get the list in standard units:

$$-1.5 \quad -0.5 \quad 0 \quad 0.5 \quad 1.5$$

Part (b). Now the average is 19, and the deviations from average are $-9, -3, 0, 3, 9$. The SD is 6. Divide by 6 to get the list in standard units:

$$-1.5 \quad -0.5 \quad 0 \quad 0.5 \quad 1.5$$

The two lists are the same in standard units.

List (b) comes from list (a) by changing the scale: multiply by 3, add 7. The 7 washes out when computing the deviations from average. The 3 washes out when dividing by the SD—because the SD got multiplied by 3 along with all the deviations. That is why the lists are the same in standard units. To summarize:

(i) Adding the same number to every entry on a list adds that constant to the average; the SD does not change.

(ii) Multiplying every entry on a list by the same positive number multiplies the average and the SD by that constant.

(iii) These changes of scale do not change the standard units.

Conversion of temperature from Fahrenheit to Celsius is a practical example:

$$C^\circ = \frac{5}{9}(F^\circ - 32^\circ)$$

Statisticians call this a *change of scale*, because it is only the units that change. (What happens if you multiply all the numbers on a list by the same negative constant? In standard units, that just reverses all the signs.)

Exercise Set F

1. A group of people have an average temperature of 98.6 degrees Fahrenheit, with an SD of 0.3 degrees.
   (a) Translate these results into degrees Celsius.
   (b) Someone's temperature is 1.5 SDs above average on the Fahrenheit scale. Convert this temperature to standard units, for an investigator who is using the Celsius scale.

The answers to these exercises are on p. A52.

7. REVIEW EXERCISES

Review exercises may cover material from previous chapters.

1. The following list of test scores has an average of 50 and an SD of 10:

39	41	47	58	65	37	37	49	56	59	62	36	48
52	64	29	44	47	49	52	53	54	72	50	50	

   (a) Use the normal approximation to estimate the number of scores within 1.25 SDs of the average.
   (b) How many scores really were within 1.25 SDs of the average?

2. You are looking at a computer printout of 100 test scores, which have been converted to standard units. The first 10 entries are

−6.2	3.5	1.2	−0.13	4.3	−5.1	−7.2	−11.3	1.8	6.3

   Does the printout look reasonable, or is something wrong with the computer?

3. From the mid-1960s to the early 1990s, there was a slow but steady decline in SAT scores. For example, take the Verbal SAT. The average in 1967 was about 543; by 1994, the average was down to about 499. However, the SD stayed close to 110. The drop in averages has a large effect on the tails of the distribution.
   (a) Estimate the percentage of students scoring over 700 in 1967.
   (b) Estimate the percentage of students scoring over 700 in 1994.

   You may assume that the histograms follow the normal curve.

   Comments. SAT scores range from 200 to 800. It does not seem that the SAT was getting harder. Most of the decline in the 1960s is thought to result from changes in the population of students taking the test. The decline in the 1970s cannot be explained that way. From 1994 to 2005, scores generally increased. The test was re-normalized in 1996, which complicates the interpretation; the averages mentioned above were converted to the new scale.[4]

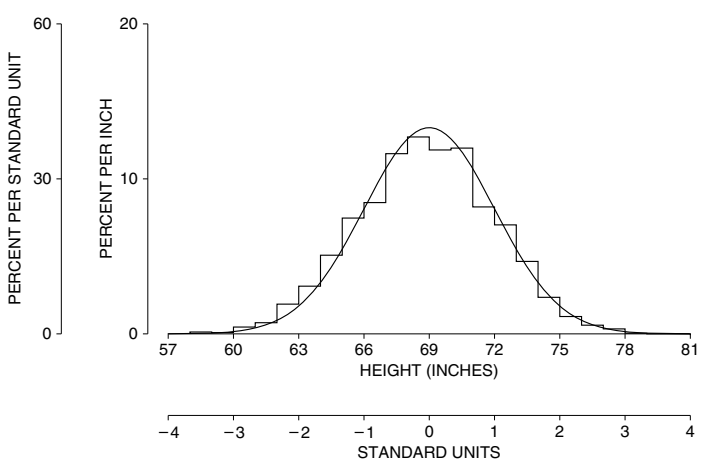
4. On the Math SAT, men have a distinct edge. In 2005, for instance, the men averaged about 538, and the women averaged about 504.
   (a) Estimate the percentage of men getting over 700 on this test in 2005.
   (b) Estimate the percentage of women getting over 700 on this test in 2005.

   You may assume (i) the histograms followed the normal curve, and (ii) both SDs were about 120.[4]

5. In HANES5, the men age 18 and over had an average height of 69 inches and an SD of 3 inches. The histogram is shown below, with a normal curve. The percentage of men with heights between 66 inches and 72 inches is exactly equal to the area between __(a)__ and __(b)__ under the __(c)__. This percentage is approximately equal to the area between __(d)__ and __(e)__ under the __(f)__. Fill in the blanks. For (a), (b), (d) and (e), your options are

   66 inches 72 inches -1 $+1$

   For (c) and (f), your options are: normal curve, histogram



6. Among applicants to one law school, the average LSAT score was about 169, the SD was about 9, and the highest score was 178. Did the LSAT scores follow the normal curve?

7. Among freshmen at a certain university, scores on the Math SAT followed the normal curve, with an average of 550 and an SD of 100. Fill in the blanks; explain briefly.
   (a) A student who scored 400 on the Math SAT was at the _____th percentile of the score distribution.
   (b) To be at the 75th percentile of the distribution, a student needed a score of about _____ points on the Math SAT.

8. True or false, and explain briefly—
   (a) If you add 7 to each entry on a list, that adds 7 to the average.
   (b) If you add 7 to each entry on a list, that adds 7 to the SD.
   (c) If you double each entry on a list, that doubles the average.
   (d) If you double each entry on a list, that doubles the SD.
   (e) If you change the sign of each entry on a list, that changes the sign of the average.
   (f) If you change the sign of each entry on a list, that changes the sign of the SD.

9. Which of the following are true? false? Explain or give examples.
   (a) The median and the average of any list are always close together.
   (b) Half of a list is always below average.
   (c) With a large, representative sample, the histogram is bound to follow the normal curve quite closely.
   (d) If two lists of numbers have exactly the same average of 50 and the same SD of 10, then the percentage of entries between 40 and 60 must be exactly the same for both lists.

10. For women age 25–34 with full time jobs, the average income in 2004 was $32,000. The SD was $26,000, and 1/4 of 1% had incomes above $150,000. Was the percentage with incomes in the range from $32,000 to $150,000 about 40%, 50%, or 60%? Choose one option and explain briefly.[5]

11. One term, about 700 Statistics 2 students at the University of California, Berkeley, were asked how many college mathematics courses they had taken, other than Statistics 2. The average number of courses was about 1.1; the SD was about 1.5. Would the histogram for the data look like (i), (ii), or (iii)? Why?

12. In 2005, the average score on the Math SAT was about 520. However, among students who took a subject-matter test, the average score on the Math SAT was about 624.[6] What accounts for the difference?

8. SUMMARY

1. The *normal curve* is symmetric about 0, and the total area under it is 100%.

2. *Standard units* say how many SDs a value is, above (+) or below (−) the average.

3. Many histograms have roughly the same shape as the normal curve.

4. If a list of numbers follows the normal curve, the percentage of entries falling in a given interval can be estimated by converting the interval to standard units, and then finding the corresponding area under the normal curve. This procedure is called the *normal approximation*.

5. A histogram which follows the normal curve can be reconstructed fairly well from its average and SD. In such cases, the average and SD are good summary statistics.

6. All histograms, whether or not they follow the normal curve, can be summarized using *percentiles*.

7. If you add the same number to every entry on a list, that constant just gets added to the average; the SD does not change. If you multiply every entry on a list by the same positive number, the average and the SD just get multiplied by that constant. (If the constant is negative, wipe out the sign before multiplying the SD.)

"LOOK, FRED! THIS SEEMS TO BE THE SAME THING, SUMMARIZED."